

TANISH SAROJ

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EDUCATION

Theem College of Engineering

Bachelor of Engineering in Computer Science | GPA: 8.0+ / 10

Mumbai, India

Expected May 2026

ACHIEVEMENTS

Winner, Smart India Hackathon (SIH) 2025 — 1st Rank Nationwide | Problem Statement #25164

Designed an autonomous drone defense system for Bharat Electronics Limited (BEL)

PROJECTS

Chronicle | Python, FastAPI, PostgreSQL, React, Docker, Prometheus

Collaborative · Lead Architect

- Designed end-to-end architecture for a prompt version control platform — immutable PostgreSQL versioning, production promotion model, full AliasHistory audit trail, and a 6-module backend (version control, execution, evaluation, synthetic data, observability, CLI) across 183 integration tests
- Architected Pareto frontier eval engine with exact-match, LLM-judge, and confidence-calibration evaluators; geometric knee-point detection surfaces plain-English deployment recommendations — e.g. "v1131 at 96.7% accuracy for \$0.000068/correct vs. 100% at 1.4× cost"
- Built robustness testing (sharp vs. flat maxima detection), a two-model synthetic data pipeline with Jaccard deduplication, Prometheus observability with webhook alerting, and a full CLI with live eval polling, colored diffs, and token-delta reporting

SimAstra – SIH 2025 National Winner | Python, PyBullet, Swarm Algorithms, NumPy

Closed Source

- Engineered decentralized Hunter-Killer algorithm enabling drone swarms to autonomously detect, prioritize, and neutralize adversarial threats without a central controller
- Implemented dynamic threat assessment logic prioritizing ground-attack drones in communication-denied environments
- Built stochastic simulation framework in PyBullet; validated optimal interception rates across 100+ combat scenarios

Insight – AI Data Analyst Agent | Python, LangChain, Streamlit, Pandas, Groq API

[Live Demo](#)

- Built conversational data analyst agent — upload a CSV, ask in plain English, receive vintage newspaper-styled visualizations and a formatted Word report; handles datasets up to 1 million rows
- Baked aesthetic consistency into the agent's system prompt, forcing Matplotlib to match the UI's paper-toned palette across all generated charts
- Implemented sandboxed code execution via LangChain Pandas Agent backed by Groq's low-latency inference

Heimdall – AI Code Review Agent | Python, FastAPI, Groq API, Llama 3 70B, Docker

[GitHub](#)

- Built agent that hooks into GitHub via webhooks and posts LLM-generated code reviews on every commit — flagging bugs, suggesting tests, and drafting documentation automatically
- Architected modular FastAPI backend with decoupled GitHubService and LLMSERVICE layers; containerized with Docker

Persona-Adaptive QA Bot | Python, PyTorch, QLoRA, Hugging Face, Phi-3

[GitHub](#)

- Fine-tuned Microsoft Phi-3-mini (3.8B) using QLoRA to dynamically adapt response style by user persona (Student vs. Expert); achieved 67% adaptation rate vs. 41% baseline on held-out test sets
- Optimized training pipeline for free Colab T4 GPU via 4-bit quantization and PEFT (zero training cost)

TECHNICAL SKILLS

Languages: Python, SQL (PostgreSQL), C++

AI / ML: PyTorch, LangChain, Hugging Face Transformers, QLoRA, PEFT, LightGBM, MCP

Backend & Infra: FastAPI, Docker, PostgreSQL, SQLAlchemy, Alembic, Prometheus, Git

Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Streamlit, PyBullet, python-docx